Reduced-Pauli entanglement checks for DLA-sector leakage mechanisms on IBM Heron hardware

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(Dated: May 25, 2026)

(Nodo DOI: 10.5281/zenodo.20382075)

Abstract

Parity-sector leakage asymmetry in small Kuramoto–XY circuits is a useful hardware observable only if the mechanism can be separated from layout, prepared-state, and readout artefacts. We report a cost-bounded reduced-Pauli entanglement/tomography check for promoted Phase 3 DLA-parity and Fisher-information-modified Kuramoto–XY circuit families. The approved `ibm_marrakesh` execution used physical qubits $[1, 2, 3, 4]$, 162 main circuits, four readout calibration circuits, 2048 shots per main circuit, and 8192 shots per readout circuit. The primary submitted IBM jobs were `d86g7h1789is738vkreg` and `d86ggpis46sc73f6v170`. A preregistered reducer produced 54 observable rows. The mean absolute measured-minus-exact deviation is 0.1299 and the maximum absolute deviation is 0.5561. The largest deviations are concentrated in transverse two-qubit correlators, especially DLA odd-sector `XX`/`YY` edge channels, while the FIM lambda-4 circuit deviates more strongly than the FIM lambda-0 reference. A tensor-product readout-mitigation replay leaves the primary mean absolute deviation essentially unchanged (0.1293), and a same-day `ibm_fez` replication gives mean absolute deviation 0.1528, readout-mitigated to 0.1513. A pinned-layout `ibm_fez` repeat with full 16-state readout calibration gives raw mean absolute deviation 0.1494; correlated readout inversion amplifies rather than erases the strongest channels, with the largest amplification still concentrated in transverse edge observables. A preregistered five-channel ZNE stress test on the same `ibm_fez` qubits gives linear-ZNE mean absolute deviation 0.4412 versus 0.4196 at scale 1, so noise scaling does not erase the dominant DLA-channel discrepancy. The FIM control channel partially improves under ZNE, suggesting that the discrepancy is structured rather than simple monotone gate noise. A third-backend `ibm_kingston` ZNE replication on qubits $[141, 142, 143, 144]$ gives the same qualitative pattern: the four DLA transverse channels move farther from exact under linear ZNE, while the FIM control improves. A same-layout Kingston statistical repeat preserves this DLA-worsens/FIM-improves separation. Larger $n = 6, 8$ reduced-Pauli/ZNE lanes on Fez and Marrakesh then address the "all hardware is $n = 4$" limitation with 36 reduced-Pauli channels per lane. A Kingston larger-width replication adds the third backend, and a Marrakesh $n = 8$ repeat checks the largest readout-sensitivity effect. The readout-mitigated linear-ZNE mean absolute deviations range from 0.0980 to 0.1577 on Fez, from 0.0604 to 0.0688 on Marrakesh after a large readout-sensitivity correction, and from 0.1504 to 1.6735 on Kingston, where the $n = 6$ full-readout assignment matrix is singular and therefore reported with a pseudo-inverse sensitivity boundary. The result is a mechanism-boundary measurement: it

shows structured reduced-Pauli distortions accompanying the small-system DLA/FIM hardware programme, but does not claim scalable tomography, quantum advantage, full-state reconstruction, or backend-general entanglement dynamics.

I. INTRODUCTION

Kuramoto synchronisation and XY spin Hamiltonians provide a compact route from phase-oscillator structure to gate-model quantum dynamics [1, 2]. The dynamical-Lie-algebra language used here follows the standard Lie-algebraic controllability framework for finite-dimensional quantum systems [3]. In the present programme, the relevant small-system hardware question is not whether a broad advantage has been demonstrated. It is narrower: when sector-resolved leakage and retention observables deviate from exact Kuramoto–XY references on IBM hardware, do reduced-Pauli correlators show an accompanying structure that can constrain the mechanism?

Previous DLA-parity and FIM runs established bounded hardware observations, including backend, layout, state-preparation, and readout sensitivity [4, 5]. A scalar leakage metric alone cannot distinguish coherent correlator deformation from population readout effects or generic depth-dependent decoherence on noisy intermediate-scale quantum hardware [6]. This paper therefore measures a small set of preregistered reduced-Pauli observables for the same class of promoted circuits. Full tomography [7] is unnecessary for this question, and the protocol is closer in spirit to targeted local-observable estimation than to full state reconstruction; related measurement-efficient ideas include classical shadows and local observable prediction [8]. The aim here is narrower still: to compare selected two-qubit correlators against exact classical references under one backend, layout, shot budget, and calibration window. The contribution is a channel-localised falsifier: the dominant DLA transverse deviations survive readout replay, backend replication, and ZNE stress tests, so future mechanism claims must account for correlator-level structure rather than only scalar leakage summaries.

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Label	Family	Initial state	Depth
`dla_even_shallow`	DLA parity	`0011`	6
`dla_odd_shallow`	DLA parity	`0001`	6
`dla_even_signal`	DLA parity	`0011`	10
`dla_odd_signal`	DLA parity	`0001`	10
`fim_lambda0_reference`	FIM pair	`0011`	4
`fim_lambda4_feedback`	FIM pair	`0011`	4

TABLE I. Promoted source circuits for the reduced-Pauli check.

II. PROTOCOL

The source circuits are listed in Table I. Four DLA-parity circuits compare even and odd initial states at shallow and signal depths. Two FIM-pair circuits compare $\lambda_{\mathrm{fim}} = 0$ and $\lambda_{\mathrm{fim}} = 4$ at fixed initial state and depth.

For each source circuit we measure nine reduced-Pauli basis settings,

$$\{\mathtt{IIXX}, \mathtt{IIYY}, \mathtt{IIZZ}, \mathtt{IXXI}, \mathtt{IYYI}, \mathtt{IZZI}, \mathtt{XXII}, \mathtt{YYII}, \mathtt{ZZII}\}. \tag{1}$$

Each source/basis pair is repeated three times at 2048 shots, giving 162 main circuits. Four computational-basis readout states, `0000`, `0001`, `0011`, and `1111`, are measured at 8192 shots. The complete hardware block therefore contains 166 circuits. The selected physical qubits are $[1, 2, 3, 4]$ on `ibm_marrakesh`; the maximum transpiled depth in the live preflight is 388, and the maximum measurement-basis depth expansion is 1.0718. A second-backend replication on `ibm_fez` used physical qubits $[21, 22, 23, 24]$, the same circuit and shot budget, and jobs `d86h4jlg7okc73el3ra0` and `d86h5d9789is738vlt7g`. A pinned-layout repeat on the same `ibm_fez` qubits added all 16 computational-basis readout calibration states and used jobs `d86hdk8p0eas73dkv9eg` and `d86hedp789is738vm7mg`. Circuit construction, transpilation, and hardware execution used Qiskit and the IBM Quantum platform [9, 10].

After the `ibm_fez` ZNE stress test, we executed a third-backend ZNE replication on `ibm_kingston`. The selected physical qubits were $[141, 142, 143, 144]$. The run used the same preregistered five channels, the same global unitary-folding scale factors $1, 3, 5$, the same three repetitions per channel and scale, 2048 main shots, and full 16-state readout calibration at 8192 shots. The completed jobs were `d86i8fas46sc73f70vg0` and

d86ijr9789is738vnh30. A same-layout statistical repeat used jobs d86pul0p0eas73dla3dg and d86pul8p0eas73dla3eg.

III. ANALYSIS

For each measured basis-rotated circuit, the reducer estimates the Pauli expectation

$$\langle P \rangle = \frac{1}{N} \sum_{b} n_b \prod_{i \in \mathrm{supp}(P)} (-1)^{b_i}, \tag{2}$$

where P is the reduced-Pauli label, n_b is the observed count for bitstring b, and N is the total shot count. Repetitions are grouped by source label and basis setting, then compared with exact reference expectations from the committed reference table data/phase3_entanglement_tomography/entanglement_observable_rows_2026-05-07.csv.

The primary output rows are committed in data/phase3_entanglement_tomography/entanglement_tomography_rows_2026-05-20.csv. The rows have SHA256 digest b280db120f9f13dfb8dd6b8d4e2e6a86a86feff714e2514be01bd5b3ed95bd83.

Readout mitigation is applied as a conservative replay, not as a stronger hardware claim. The four calibration circuits are marginalised into independent single-qubit assignment matrices in IBM count-key order, and the tensor-product assignment matrix is inverted before recomputing each reduced-Pauli expectation, following the standard response-matrix/post-processing measurement-error mitigation logic [11]. This is not a full 16-state correlated readout calibration and is not ZNE or PEC. When all 16 calibration states are present in the pinned ibm_fez repeat, the reducer instead inverts the full correlated 16-by-16 assignment matrix. This still corrects only readout assignment; it does not correct basis rotations, coherent gate errors, drift, or depth effects.

IV. RESULTS

The approved IBM execution completed on ibm_marrakesh. The submitted jobs were d86g7h1789is738vkreg and d86ggpis46sc73f6v170. The analysis produced 54 observable rows with mean absolute deviation 0.1299 and maximum absolute deviation 0.5561 relative to exact references.

Circuit label	Mean signed	Mean absolute	Max absolute
`dla_even_shallow`	-0.120474	0.148396	0.314180
`dla_even_signal`	0.061095	0.063458	0.160443
`dla_odd_shallow`	0.098416	0.219708	0.515473
`dla_odd_signal`	0.153497	0.176026	0.556091
`fim_lambda0_reference`	0.030496	0.052205	0.088148
`fim_lambda4_feedback`	0.096107	0.119566	0.274155

TABLE II. Label-level measured-minus-exact deviation aggregates. Each row summarises nine reduced-Pauli observables.

Table II gives the label-level aggregates. DLA circuits have larger mean absolute deviation than the FIM pair overall. Within the DLA family, the odd shallow and odd signal circuits produce the largest aggregate departures. Within the FIM pair, $\lambda_{\text{fim}} = 4$ has more than twice the mean absolute deviation of $\lambda_{\text{fim}} = 0$.

Figure 1 shows the signed measured-minus-exact deviations by source label and basis setting. The strongest pattern is not a uniform decay of all correlators. Instead, deviations concentrate in transverse two-qubit channels. The odd-sector DLA signal circuit has large positive shifts in `XXII` and `YYII`; the odd shallow circuit has analogous positive shifts in `IIXX` and `IIYY`. The even shallow circuit moves in the opposite direction on corresponding transverse channels.

The largest individual deviations are shown in Table III. The top four are DLA odd-sector transverse correlators. The strongest FIM-channel departure is the `IXXI` observable for the $\lambda_{\text{fim}} = 4$ circuit.

A. Readout mitigation and same-day replication

Table IV summarises the primary `ibm_marrakesh` run, the same-day `ibm_fez` replication, and the pinned-layout `ibm_fez` repeat with full readout calibration. The tensor-product readout inversion does not collapse the deviations toward zero on either backend. On `ibm_marrakesh`, the mean absolute deviation changes from 0.1299 to 0.1293; on `ibm_fez`, it changes from 0.1528 to 0.1513. The pinned-layout repeat gives a similar raw mean absolute

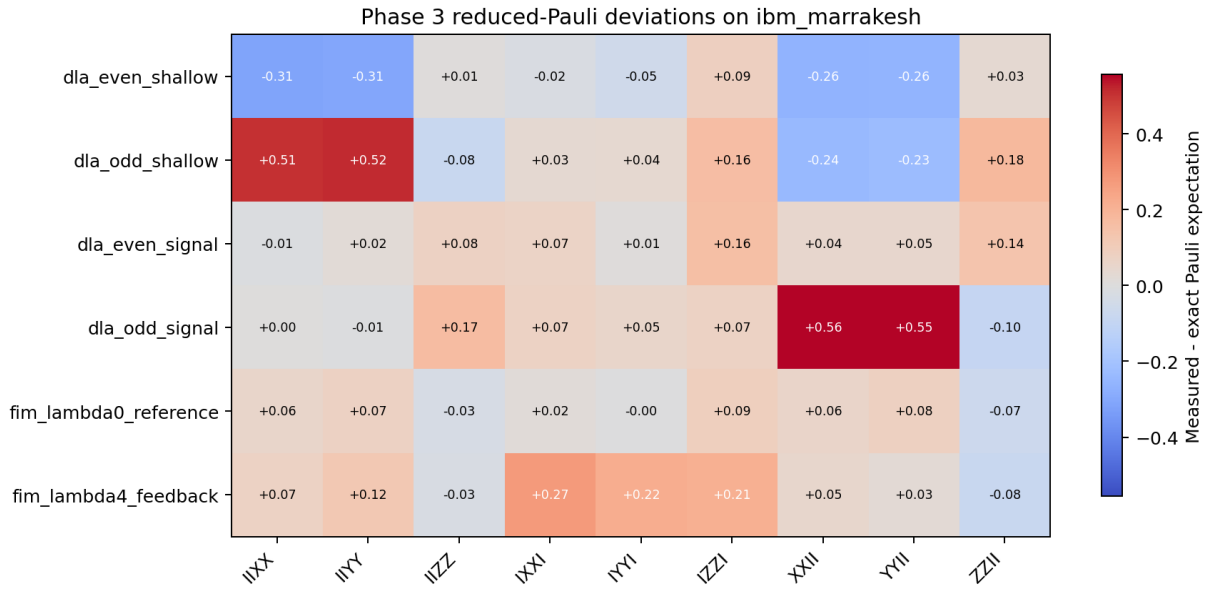


FIG. 1. Measured-minus-exact reduced-Pauli deviations for the completed `ibm_marrakesh` run. The strongest deviations concentrate in transverse edge correlators rather than in all observables uniformly.

Circuit label	Basis	Measured	Exact	Deviation
`dla_odd_signal`	`XXII`	0.431641	-0.124450	0.556091
`dla_odd_signal`	`YYII`	0.429688	-0.124450	0.554138
`dla_odd_shallow`	`IIYY`	0.488932	-0.026541	0.515473
`dla_odd_shallow`	`IIXX`	0.478841	-0.026541	0.505382
`dla_even_shallow`	`IIXX`	-0.456055	-0.141874	-0.314180
`dla_even_shallow`	`IIYY`	-0.447591	-0.141874	-0.305717
`fim_lambda4_feedback`	`IXXI`	0.175781	-0.098373	0.274155

TABLE III. Largest measured-minus-exact deviations in the reduced-Pauli rows.

deviation, 0.1494, while full correlated readout inversion increases the aggregate mitigated deviation to 0.1953 and the largest mitigated deviation to 0.9913. This does not prove backend-general dynamics, but it does rule out the simplest interpretation that the primary observation is only a single-backend population-readout artefact. The full correlated inversion result is treated as a sensitivity bound, because matrix inversion can amplify calibration and shot noise in individual channels.

Backend	Mean abs.	Max abs.	Mitigated mean abs.	Mitigated max abs.
`ibm_marrakesh`	0.129893	0.556091	0.129304	0.564834
`ibm_fez` auto layout	0.152821	0.479015	0.151317	0.486875
`ibm_fez` pinned full readout	0.149445	0.473481	0.195282	0.991278

TABLE IV. Cross-backend reduced-Pauli deviation summary. Mitigation is independent single-qubit readout inversion for the first two rows and full correlated readout inversion for the pinned full-readout row.

The amplification is not uniform across the reduced-Pauli table. In the pinned full-readout replay, transverse edge channels have mean absolute amplification 0.1110 and maximum amplification 0.5178, while transverse-middle channels have mean absolute amplification 0.0026. The two largest amplified rows are `dla_odd_shallow/IIYY` and `dla_odd_shallow/IIXX`, both already in the dominant DLA transverse-edge family. The `ZZ` edge class also shows sensitivity, with mean absolute amplification 0.0946 and maximum amplification 0.2855. Thus the correlated-inversion replay should not be read as a clean mitigation result, but it also does not look like a featureless uniform noise blow-up. The supplementary amplification rows are generated in `data/phase3_entanglement_tomography/entanglement_tomography_full_readout_amplification_top_2026-05-20.csv` and `data/phase3_entanglement_tomography/entanglement_tomography_full_readout_amplification_classes_2026-05-20.csv`.

B. ZNE stress test

After the full-readout replication, we executed the preregistered ZNE subset proposed by the mechanism-boundary analysis. The run used `ibm_fez` qubits $[21, 22, 23, 24]$, global unitary-folding scale factors $1, 3, 5$, three repetitions per channel and scale, and the full 16-state readout calibration. This follows the short-depth error-mitigation and digital ZNE framework [12–15]. The submitted jobs were `d86hs6qs46sc73f70h90` and `d86hsltg7okc73el4lg0`. The resulting artefact contains 45 main circuits and 16 readout-calibration circuits. The ZNE reducer groups rows by channel and noise scale, computes raw and full-correlated-readout-mitigated expectations, and performs linear and quadratic extrapolations to zero noise.

Circuit label	Basis	Scale-1 abs.	Linear ZNE abs.	Mitigated linear abs.
dla_odd_shallow	IIXX	0.490083	0.524344	0.535976
dla_odd_shallow	IIYY	0.457531	0.493067	0.503956
dla_odd_signal	XXII	0.340596	0.378709	0.386027
dla_odd_signal	YYII	0.359151	0.393086	0.400958
fim_lambda4_feedback	IZZI	0.450778	0.417032	0.407170

TABLE V. Preregistered ZNE stress-test summary on ibm_fez. The table reports absolute deviation from exact reference at scale 1, after linear extrapolation from scale factors $1, 3, 5$, and after full-correlated readout inversion followed by linear extrapolation.

Table V gives the channel-level summary. The aggregate result is a negative mitigation outcome for the DLA channels: the scale-1 mean absolute deviation across the five selected channels is 0.4196, while linear ZNE gives 0.4412 and full-readout-mitigated linear ZNE gives 0.4468. The four DLA transverse channels all move farther from the exact reference under linear ZNE. The FIM control behaves differently: fim_lambda4_feedback/ IZZI improves from scale-1 deviation 0.4508 to linear-ZNE deviation 0.4170, and the quadratic fit gives 0.2294. Because quadratic extrapolation with three scales can overfit the three points, it is reported as a sensitivity check rather than a stronger correction claim.

The third-backend ibm_kingston replication gives the same qualitative channel separation, as shown in Table VI. The scale-1 mean absolute deviation is 0.4534, linear ZNE gives 0.4729, and full-readout-mitigated linear ZNE gives 0.4806. All four DLA transverse channels again move farther from exact under linear ZNE. The FIM control moves in the opposite direction, improving from scale-1 deviation 0.2490 to linear-ZNE deviation 0.1673 and quadratic-ZNE deviation 0.0490. This is not a backend-general dynamics claim, but it is a third-backend qualitative replication of the Fez stress-test pattern under the same five-channel protocol. The same-layout Kingston repeat preserves the pattern: scale-1 mean absolute deviation is 0.4656, linear ZNE gives 0.4816, and full-readout-mitigated linear ZNE gives 0.4893. In the repeat, all four DLA transverse channels again move farther from exact under linear ZNE, while the FIM control improves from scale-1 deviation 0.2564 to linear-ZNE deviation 0.1941.

Circuit label	Basis	Scale-1 abs.	Linear ZNE abs.	Mitigated linear abs.
dla_odd_shallow	IIXX	0.501476	0.525673	0.541006
dla_odd_shallow	IIYY	0.508963	0.546913	0.563012
dla_odd_signal	XXII	0.504333	0.571200	0.583571
dla_odd_signal	YYII	0.503031	0.553459	0.565459
fim_lambda4_feedback	IZZI	0.248955	0.167331	0.149998

TABLE VI. Third-backend ZNE replication on ibm_kingston. The five-channel protocol is identical to Table V; all four DLA transverse channels worsen under linear ZNE, while the FIM control improves.

C. Larger-width reduced-Pauli stress lanes

To avoid leaving the hardware package entirely at $n = 4$, we executed a bounded larger-width extension on ibm_fez and ibm_marrakesh at $n = 6$ and $n = 8$. The larger lanes retain the reduced-Pauli philosophy rather than attempting scalable tomography: six source families, left/middle/right transverse edge XX/YY observables, local-folding ZNE scales $1, 3, 5$, three repetitions per channel and scale, and complete 2^n computational-basis readout calibration. The submitted jobs were d870qa8p0eas73dlj6kg, d870qf9789is73909nu0, d870qj9789is73909o2g, and d870qo0p0eas73dlj75g. A dedicated dynamic-width reducer generated exact reference rows, raw scale rows, full-readout-mitigated rows, and linear/quadratic ZNE channel summaries.

Table VII summarises the aggregate result. The larger-width lanes are not a simple monotone extrapolation of the $n = 4$ story. Fez remains moderate in raw and mitigated deviation. Marrakesh shows much larger raw scale-1 and raw linear-ZNE deviations but a strong reduction after full readout inversion, and the same pattern is reproduced in the Marrakesh $n = 8$ repeat. Kingston provides the third larger-width backend: the $n = 8$ lane is finite after full readout inversion, while the $n = 6$ lane has a singular full-readout assignment matrix and is therefore reduced with a pseudo-inverse sensitivity boundary. This keeps the claim boundary honest: the $n = 6, 8$ lanes show that reduced-Pauli structure can be measured beyond four qubits with archived raw counts and complete readout calibration, but they also show that larger-width conclusions are readout-sensitive and backend/layout

Backend	n	Scale-1 mean abs.	Linear ZNE mean abs.	Mitigated linear mean abs.
ibm_fez	6	0.1697	0.1688	0.1577
ibm_fez	8	0.1348	0.1395	0.0980
ibm_marrakesh	6	0.3306	0.3298	0.0688
ibm_marrakesh	8	0.3788	0.3795	0.0604
ibm_marrakesh	8 repeat	0.3917	0.3906	0.0609
ibm_kingston	6	0.2897	0.2916	1.6735
ibm_kingston	8	0.4710	0.4704	0.1504

TABLE VII. Phase 3 larger-width reduced-Pauli/ZNE stress lanes. Each row summarises 36 reduced-Pauli channels from the retrieved 2026-05-20 IBM raw-count artefacts. The mitigated column applies full correlated computational-basis readout inversion before the linear ZNE fit, except for the Kingston $n = 6$ row where the readout assignment matrix was singular and a pseudo-inverse sensitivity replay is reported.

dependent.

V. DISCUSSION

The main result is not that the hardware reproduces the exact reduced-Pauli reference structure. It does not. The contribution is that the departures from the exact reference are large, structured, and concentrated in specific correlator channels. This turns the earlier leakage/retention observations into a sharper hardware-mechanism question: the same circuit families that show sector-sensitive survival behaviour also show measurable correlator-level distortions.

The DLA circuits dominate the strongest deviations. The sign structure is particularly important: odd-sector circuits show large positive transverse edge-correlator shifts, whereas the even shallow circuit shows large negative shifts on corresponding channels. This is not consistent with a single scalar decoherence factor multiplying every observable equally. It is more naturally read as a sector- and edge-dependent deformation under the selected backend, layout, circuit family, and calibration window.

The FIM result is more muted but still useful. The $\lambda_{\mathrm{fim}} = 4$ feedback circuit deviates

DLA odd channels	Dominant transverse edge deviations in `XX`/`YY` rows.
DLA even channels	Opposite-sign shallow transverse structure, not a scalar decoherence rescaling.
FIM control	Smaller but persistent $\lambda_{\text{fim}} = 4$ distortion, with partial ZNE improvement.
Readout boundary	Marrakesh larger-width rows are strongly readout-sensitive; Kingston $n = 6$ requires a pseudo-inverse replay.

FIG. 2. Reduced-Pauli channel map distilled from the detailed heatmap and tables. The useful signal is localisation by channel family and mitigation response, not a claim of full-state tomography.

more strongly than the $\lambda_{\text{fim}} = 0$ reference. This is consistent with the earlier negative fixed-FIM hardware result: the tested digital FIM modification does not simply protect the target structure and introduces a measurable reduced-Pauli distortion channel. The result motivates adaptive FIM follow-up only under a conservative claim boundary; it does not rescue a fixed-λ_{fim} protection claim.

The readout boundary remains central. Four readout calibration states are included in each execution block, and independent single-qubit readout inversion leaves the aggregate deviation scale intact. Because the largest deviations occur in transverse basis-rotated measurements, not only in computational-basis `ZZ` channels, the result is unlikely to be a pure population-readout story. It may still include basis-rotation error, layout-dependent coherent error, calibration drift, and depth-dependent decoherence. The safe interpretation is therefore mechanism-boundary evidence, not full causal attribution.

The two `ibm_fez` executions also separate layout robustness from same-layout repeatability. The auto-layout `ibm_fez` run and the pinned-layout repeat use the same physical qubits, $[21, 22, 23, 24]$, and their raw aggregate deviations are close: 0.1528 and 0.1494. The remaining quantitative differences may reflect calibration-window drift, basis-rotation error, coherent layout-specific error, or finite-shot variation. Because the same-order transverse deviations persist across `ibm_marrakesh` and `ibm_fez`, and across two `ibm_fez` executions, a single layout-only explanation is not sufficient; however, layout-specific coherent error

remains a plausible contributor to the detailed channel amplitudes.

The ZNE stress test sharpens this conclusion. If the dominant DLA transverse deviations were mainly a monotone gate-noise amplitude effect, the folded-scale trend would be expected to extrapolate back toward the exact reference. It does not for the four preregistered DLA channels. Instead, the linear ZNE estimate is farther from exact than the scale-1 value. The FIM control shows a partial correction in the opposite direction, so the negative DLA result is not merely a universal failure of the fitting machinery. The safer reading is that the DLA transverse discrepancy is structured under this hardware window and not removed by this simple global-folding ZNE lane.

The `ibm_kingston` replication strengthens this point without changing the claim boundary. It repeats the same DLA-worsens/FIM-improves separation on a third accessible Heron backend, using a different physical layout from the `ibm_fez` run. The channel amplitudes are not identical, as expected for finite shots, different calibration windows, and layout-dependent coherent error, but the sign of the ZNE stress-test conclusion is preserved. The same-layout Kingston repeat strengthens the statistical reading: under the same backend, physical qubits, channels, scales, and readout-calibration design, the DLA-worsens/FIM-improves separation recurs. This remains a same-layout drift-robustness statement, not an inference of backend-general entanglement dynamics.

A. Readout-sensitivity check

The four readout calibration circuits provide a scale check for the largest reduced-Pauli deviations. The dominant measured bitstring for each calibration state is shown in Table VIII. Because Qiskit reports classical bitstrings with endianness opposite to the source labels for some prepared states, the table reports the dominant observed bitstring rather than treating the printed label as a direct left-to-right comparator. The mean dominant-state retention across the four calibration circuits is 0.9771, corresponding to a mean leakage from the dominant calibration bitstring of 0.0229. The weakest calibration state retains 0.9744 of shots.

This calibration block does not remove the need for caution: transverse basis-rotated measurements can be affected by basis-rotation and coherent gate errors before readout. It does, however, make a pure computational-basis readout-retention explanation implausible

Prepared label	Dominant observed bitstring	Retention
0000	0000	0.981079
0001	1000	0.975342
0011	1100	0.977661
1111	1111	0.974365

TABLE VIII. Readout calibration dominant-state retention for the completed `ibm_marrakesh` execution. The largest reduced-Pauli deviations in Table III are roughly an order of magnitude larger than this dominant-bitstring readout leakage scale.

for the largest observed effects. The maximum reduced-Pauli deviation, 0.5561, is about 21.7 times the worst dominant-state calibration leakage of 0.0256 and about 24.3 times the mean dominant-state calibration leakage of 0.0229. The paper's claim boundary is therefore refined as follows: readout calibration does not by itself explain the largest transverse correlator deviations, but the result remains a compound hardware-window observation rather than an isolated entanglement mechanism attribution.

VI. CLAIM BOUNDARY

The supported claims are intentionally narrow. The run measured preregistered reduced-Pauli correlators for small Kuramoto–XY DLA and FIM circuit families on one IBM Heron backend and layout. The measured correlators deviate from exact references in a structured way, with the largest deviations in transverse edge channels. This constrains the mechanism interpretation of the earlier DLA/FIM hardware programme.

The result does not support quantum advantage, scalable tomography, full-state reconstruction, or claims about unmeasured circuits and depths. The `ibm_fez` and `ibm_kingston` ZNE replications support limited multi-backend robustness for the five-channel stress-test pattern, not backend-general entanglement dynamics.

The larger-width $n = 6, 8$ lanes further show that reduced-Pauli stress tests can be executed beyond four qubits, including a third-backend Kingston replication. The Marrakesh larger-width readout correction is large, and the Kingston $n = 6$ calibration matrix is singular. These rows must therefore be treated as readout-sensitive stress evidence rather than a

clean backend-general scaling claim.

The Kingston $n = 6$ singular matrix is itself informative about the boundary of full correlated mitigation at larger width. The calibration block contains 2^6 prepared computational-basis states, but finite shots, classical readout-key collisions, and near-degenerate assignment columns can make the empirical matrix rank deficient. In that case direct inversion would turn a calibration artefact into a numerical artefact, so the reducer reports a pseudo-inverse sensitivity replay and the manuscript does not treat that row as ordinary full-readout mitigation.

At the time of execution, authenticated live QPU access for this paper was limited to IBM Quantum hardware and to the reported Heron backends. The repository contains provider-neutral execution and artefact-packaging surfaces for other gate-model, trapped-ion, neutral-atom, photonic, annealing, analogue, and simulator targets, but those software routes are not evidence of live non-IBM execution. We plan to rerun the preregistered reduced-Pauli and ZNE stress-test payloads on additional hardware families when compute allocations and provider access become available; until then, the claim remains an IBM-Heron hardware-window result rather than a cross-provider mechanism claim.

VII. REPRODUCIBILITY

The source code and data are in the anulum/scpn-quantum-control repository. The execution and analysis use Qiskit and IBM Quantum access paths consistent with the cited platform references [9, 10]. The relevant commands are:

- live readiness and gated submission: `scripts/phase3_entanglement_tomography_ibm.py`;
- completed raw-count artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_marrakesh_2026-05-20T004334Z.json`;
- completed replication raw-count artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_fez_2026-05-20T014536Z.json`;
- completed pinned full-readout raw-count artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_fez_2026-05-20T020452Z.json`;

- post-run reducer: `scripts/analyse_phase3_entanglement_tomography.py`;
- paper-asset generator: `scripts/generate_phase3_entanglement_paper_assets.py`.
- ZNE-subset readiness artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_fez_2026-05-20T023204Z.json`.
- completed ZNE-subset raw-count artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_fez_2026-05-20T023600Z.json`;
- completed Kingston ZNE raw-count artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_kingston_2026-05-20T030211Z.json`;
- Kingston ZNE reducer summary: `data/phase3_entanglement_tomography/entanglement_zne_summary_2026-05-20_ibm_kingston_zne.json`;
- completed Kingston same-layout statistical repeat artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_kingston_2026-05-20T114719Z.json`;
- Kingston same-layout repeat reducer summary: `data/phase3_entanglement_tomography/entanglement_zne_summary_2026-05-20_ibm_kingston_zne_repeat.json`;
- ZNE reducer: `scripts/analyse_phase3_entanglement_zne.py`;
- larger-width Phase 3 analysis summaries: `data/phase3_entanglement_tomography/phase3_large_system_analysis_20260520T195233Z_ibm_fez_n6.json`, `data/phase3_entanglement_tomography/phase3_large_system_analysis_20260520T195245Z_ibm_fez_n8.json`, `data/phase3_entanglement_tomography/phase3_large_system_analysis_20260520T195300Z_ibm_marrakesh_n6.json`, and `data/phase3_entanglement_tomography/phase3_large_system_analysis_20260520T195312Z_ibm_marrakesh_n8.json`;
- additional larger-width variant summaries: `data/phase3_entanglement_tomography/phase3_large_system_analysis_20260520T201613Z_ibm_kingston_n6.json`, `data/phase3_entanglement_tomography/phase3_large_system_analysis_20260`

520T201657Z_ibm_marrakesh_n8.json, and data/phase3_entanglement_tomography/phase3_large_system_analysis_20260520T201817Z_ibm_kingston_n8.json;

- larger-width retrieval/reducer: scripts/retrieve_phase3_large_system_ibm.py.

VIII. CONCLUSION

A preregistered 166-circuit IBM Heron run measured 54 reduced-Pauli observables for promoted DLA and FIM Kuramoto–XY circuits. The resulting deviations from exact references are large and structured, with the strongest departures in transverse DLA edge correlators and a smaller but clear FIM $\lambda_{\mathrm{fim}} = 4$ distortion. The paper's contribution is a bounded mechanism-separation result: it localises correlator-level hardware deformation accompanying the earlier leakage/retention observations, shows that the largest effects exceed the dominant computational-basis readout leakage scale, and verifies that independent readout inversion and a second Heron backend do not remove the aggregate deviation pattern. A pinned-layout repeat with full readout calibration further shows that correlated readout inversion amplifies rather than eliminates the strongest deviations. The completed ZNE stress test adds a stronger falsifier: simple global-folding ZNE does not erase the four dominant DLA transverse deviations, even though the FIM control partially improves. A third-backend ibm_kingston ZNE replication preserves the same qualitative channel separation, and a same-layout statistical repeat preserves it again as a drift-robustness check. The present paper therefore remains narrow but stronger: it reports structured mechanism-boundary evidence that survives readout replay, second-backend replication, pinned-layout repetition, third-backend ZNE replication, and a small controlled noise-scaling stress test, plus larger-width $n = 6, 8$ reduced-Pauli/ZNE lanes. It still avoids scalable tomography, backend-general, full-causal-entanglement, or advantage claims.

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